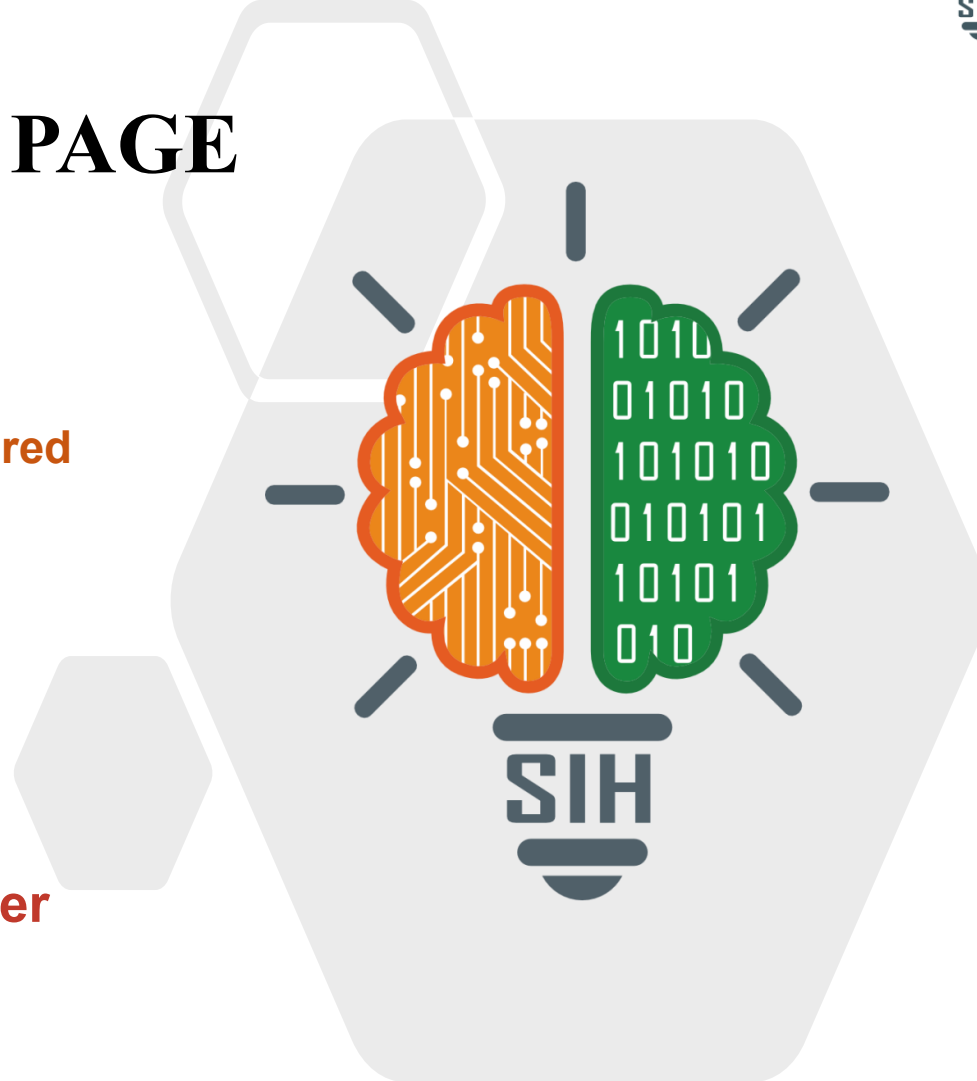


TITLE PAGE

- Problem Statement ID – **26121**
- Problem Statement Title – **eRTMAC-NWIS**
(Nearby Wells Intelligence System): An AI-Powered
Offset Well Knowledge and Decision Support
Platform for Drilling Operations
- Theme – **Smart Automation**
- PS Category – **Software**
- Team ID – **[Enter Team ID]**
- Team Name (Registered on portal) – **[Enter
Team Name]**



NWIS: Nearby Wells Intelligence System

Proposed Solution (Describe your Idea/Solution/Prototype)

- **Problem** – offset-well knowledge is scattered across well completion reports (WCR), daily drilling reports (DDR), mud logs, scans and memory; the nearest well is not always the relevant one; past trouble surfaces too late
- **Idea** – NWIS, a standalone AI/ML decision-support platform beside eRTMAC, turns those records into a searchable knowledge base and warns the driller before the bit reaches trouble seen in nearby wells
- **Prototype** – working end to end on a synthetic Upper Assam pilot: 61 wells, 10 fields, 180 reports, 7-screen dashboard

Detailed explanation of the proposed solution

- **Read** – OCR + NLP pull every event from the reports: depth, formation, severity, lost time (NPT), remedial action, lessons
- **Remember** – one knowledge base with events, lessons, casing, cement jobs, reservoir data; one search box over every event, lesson and report line
- **Rank** – nearby wells on a map within a user-defined radius, scored on 7 factors, not distance alone
- **Correlate** – formation tops tie each offset to this well's depth scale
- **Warn** – look-ahead alerts for the next 300 m: risk, confidence, reasons, evidence, recommended actions
- **Predict** – six trained hazard models and a mud-weight check for overpressure

How it addresses the problem

- **Map** – offset wells around the active well within a chosen radius
- **Instant access** – offset events, lessons and mitigations, one click from the source report page
- **Correlation** – mud losses, kicks, stuck pipe and cementing projected across wells by depth and formation; drilling parameters compared per formation; casing, cement and reservoir records for every well
- **Proactive alerts** – before the bit reaches past trouble, refreshed as the bit advances
- **Prediction** – mud loss, stuck pipe, kick, instability, torque & drag, cementing issues

Innovation and uniqueness of the solution

- **Relevance, not proximity** – explained 7-factor ranking engineers can re-weight
- **Formation-tie correlation** – a loss at 2,840 m in an offset is placed where this well meets the same rock
- **Evidence-first** – every offset event traces to document → page → line
- **Two views** – evidence score and trained-model probability, side by side
- **One vote per well** – an incident in WCR, DDR and mud log counts once, all citations kept
- **Crew-proven actions first** – recommendations lead with what offset crews actually did

Technologies to be used

- **Frontend** – React 18, Vite, Tailwind CSS, Leaflet maps (OpenStreetMap)
- **Backend / API** – Python 3.11+, FastAPI, Uvicorn, Server-Sent Events
- **Data & search** – SQLite with FTS5 (BM25); PostgreSQL + PostGIS in production
- **AI / ML / NLP** – rule-based NLP extraction, logistic-regression hazard models, OCR (Tesseract + PyMuPDF)
- **Analytics** – geo-distance, formation-tie correlation, trend detection, hazard clustering
- **DevOps** – Docker, Git/GitHub, 39 automated tests
- **Integration** – WITSML / eRTMAC connector (simulated feed in prototype)

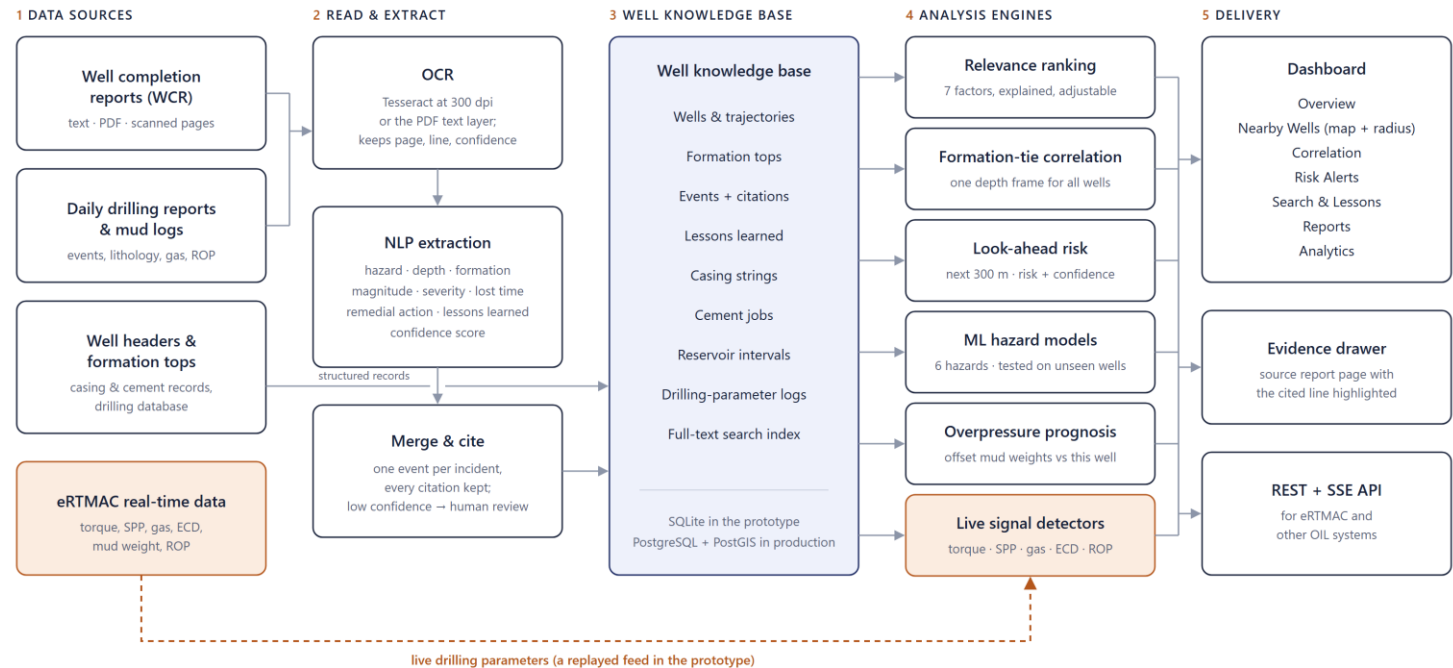
How NWIS decides

- **Relevance** – geology 26%, proximity 18%, experience 15%, depth 14%, parameters 12%, trajectory 10%, recency 5%
- **Risk** – recurrence × severity × depth agreement; confidence shown separately
- **Models** – leave-one-well-out AUC 0.82–0.93 (synthetic pilot)

Methodology and process for implementation (Flow Charts/Images/ working prototype)

NWIS architecture

From scattered drilling reports to proactive, evidence-backed alerts



Every alert keeps its evidence trail: report → page → line. The prototype runs on a synthetic Upper Assam dataset.

Working prototype: github.com/yassudicoder/eRTMAC-NWIS · reports → OCR/NLP → cited events → knowledge base → ranked offsets → formation-tie correlation → look-ahead risk + ML → dashboard and REST/SSE API for eRTMAC

Analysis of the feasibility of the idea

- **Works today** – reports to cited alerts end to end; 7-screen dashboard, REST API, 39 passing tests
- **Pilot data** – synthetic Upper Assam: 61 wells, 10 fields, 180 reports; 1,284 mentions merged into 547 cited events
- **Measured** – extraction precision 100%, recall 99.1%; model AUC 0.82–0.93 on unseen wells (synthetic data; real accuracy measured in the pilot)
- **Data exists** – OIL has 3,000+ wells of history [1]; WCRs, DDRs, mud logs and eRTMAC feeds are recorded today
- **Proven payoff** – offset-driven planning saved 15 rig days on a deep well in NE India [3]
- **Low cost, sustainable** – open-source stack on OIL's own servers; OIL owns the knowledge base; models retrain as wells are drilled
- **Roadmap** – Prototype (done) → Pilot on one Upper Assam structure with real data + eRTMAC (3–6 months) → all OIL areas

Potential challenges and risks

Strategies for overcoming these challenges

Scattered, inconsistent, scanned records

OCR + NLP with confidence scores; low-confidence items flagged for review

Nearest well may not be relevant

Explained 7-factor relevance with adjustable weights

Same formation at different depths

Formation-tie correlation with a tie-quality score

Wrong or invented AI output

Every event cited to page and line; confidence shown apart from risk; model accuracy published

No live eRTMAC access in the prototype

Simulated real-time connector; swap in a WITSML / eRTMAC client

Proprietary, sensitive data

Synthetic data for the prototype; on-premise deployment with role-based access and encryption

Few offsets drilled a formation

Each alert states how many wells support it; single-well evidence is flagged

Potential impact on the target audience

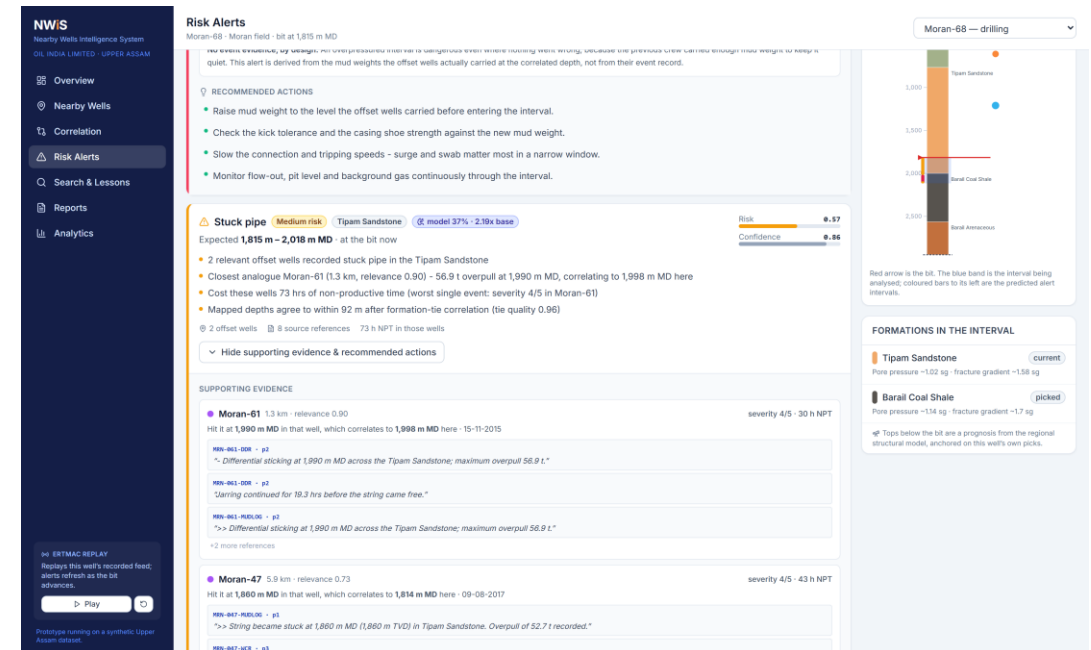
- **Drilling engineers** – relevant offset experience and risk warnings in seconds, before the bit reaches the interval
- **Geologists** – automatic cross-well correlation and formation-wise event history
- **Rig & operations teams** – fewer surprises, crew-proven mitigation steps, less non-productive time
- **OIL management** – recurring hazard zones and lost time by hazard and formation; knowledge retained when people move on

Why it matters

- In the Netherlands, an estimated 20% of drilling time goes to recovering from unexpected, potentially avoidable incidents [4]
- Stuck pipe and borehole problems: over 24% of NPT in one operator's shaly formations [5]; lost circulation about 12% of NPT, USD 1–2 billion a year [6][7]
- Overpressure mapped in the Barail and Kopili of the upper Assam Shelf [8]

Benefits of the solution (social, economic, environmental, etc.)

- **Economic** – cuts avoidable NPT; better casing, mud and cement design from past lessons; lower well cost
- **Safety & social** – earlier warning of well-control risk; knowledge shared across teams and generations
- **Environmental** – fewer well-control incidents, less drilling fluid lost to formations, fewer sidetracks and re-drills
- **National** – India imported 88.7% of its crude oil in 2025-26 (provisional) [9]; safer, faster domestic drilling supports energy security, Atmanirbhar Bharat and Viksit Bharat 2047



Prototype: a stuck-pipe alert with offset evidence and the model's second opinion (synthetic pilot data)

Details / Links of the reference and research work

- [1] Oil India Limited, Drilling (official website) – oil-india.com/drilling
- [2] Oil India Limited, Integrated Annual Report 2025-26 – oil-india.com (financial results)
- [3] Bhardwaj et al., Deep wells aided by geomechanics and real-time decisions, NE India, SPE-200627-MS, 2020 – onepetro.org
- [4] Hoetz, Jaarsma & Kortekaas, Drilling Hazards Inventory, SPE-166254-MS, 2013 – onepetro.org
- [5] Aljubran, Al-Yami & Madan, Preventing stuck pipe in offshore shaly formations, SPE-183717-MS, 2017 – onepetro.org
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- [8] Rai, Singha & Chatterjee, 3D pore pressure and overpressure, upper Assam Shelf, Acta Geophysica 70, 2022 – doi.org/10.1007/s11600-022-00771-1
- [9] PPAC (MoPNG), India's Oil & Gas Ready Reckoner FY 2025-26 – ppac.gov.in
- [10] Krygier et al., NPT due to well stability issues (93 wells), SPE-200732-MS, 2020 – onepetro.org
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- [12] Parmar et al., Risk assessment of offset wells using cognitive analysis, IPTC-24705-MS, 2025 – onepetro.org
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- [14] Guillot et al., ML prediction of lost circulation at well planning, OTC-34764-MS, 2024 – onepetro.org
- [15] DGH National Data Repository, Assam Arakan Basin – ndrdgh.gov.in
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- [17] Energistics, WITSML data standards – energistics.org/witsml-data-standards
- [18] Docs: FastAPI (fastapi.tiangolo.com), SQLite FTS5 (sqlite.org/fts5.html), Tesseract OCR, PostGIS ST_DWithin, Leaflet (leafletjs.com)

Prototype & code

github.com/yassudicoder/eRTMAC-NWIS · FastAPI + React dashboard · synthetic Upper Assam dataset (no OIL data used) · 39 automated tests